

SUFFICIENCY ENABLING INFRASTRUCTURE AND STRUCTURAL CHANGE UNDER A CARBON BUDGET

Doctoriales 2026

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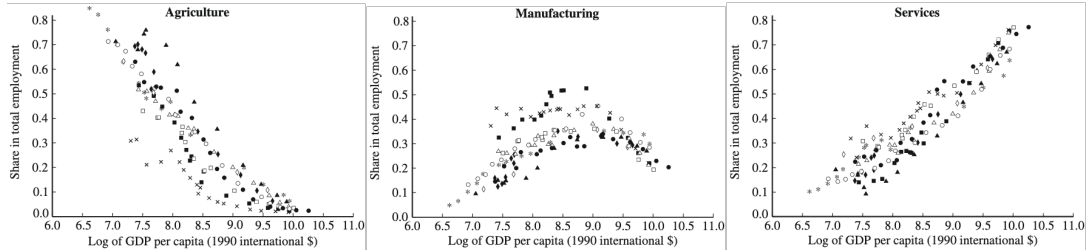


June 09. 2026



1. INTRODUCTION

STRUCTURAL CHANGE



► *Sectoral composition (Herrendorf et al., 2014)*

SECTORAL COMPOSITION

EMPLOYMENT DYNAMICS

INDUSTRIAL ORGANIZATION

TECHNICAL CHANGE

DEMAND

INSTITUTIONS



RESEARCH QUESTION

Low-demand transitions should be seen as a process of directed structural change constrained by physical limits



This paper

Multi-sector growth model

1

Infrastructure shapes preferences endogenously

2

Binding carbon budget

3



2. MODEL

MODEL

$$Y_i = A_i L_i$$

$$\dot{S} = A_S L_S - \delta_S S$$

$$\dot{E} = e_b c_b + e_S c_S$$

Sufficiency enabling infrastructure

Depreciation

Emissions

Carbon intensity

Non-homothetic CES preferences:

$$\Omega_b(S)^{1/\sigma} \left(\frac{c_b}{\psi^{\varepsilon_b}} \right)^{\frac{\sigma-1}{\sigma}} + \Omega_g^{1/\sigma} \left(\frac{c_g}{\psi^{\varepsilon_g}} \right)^{\frac{\sigma-1}{\sigma}} = 1$$

Elasticity of substitution

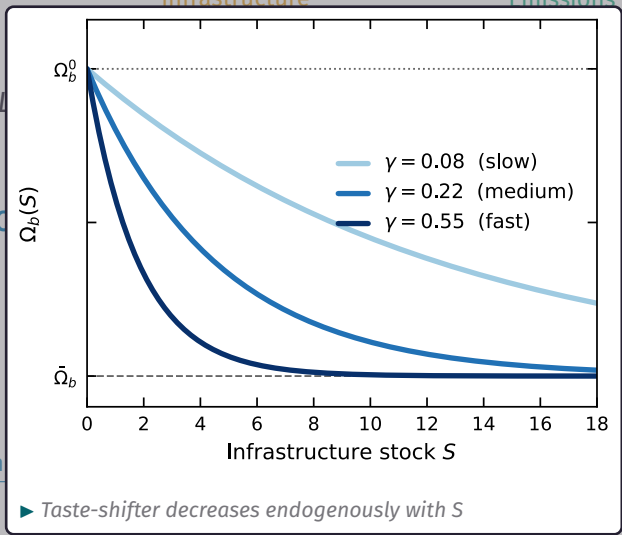
Consumption

Taste shifter

Utility

Non-homotheticity parameter

MODEL



PLANNER'S PROBLEM

A planner maximizes intertemporal welfare :

$$\int_0^T e^{-\rho t} \frac{\psi^{1-\theta} - 1}{1-\theta} dt$$

subject to

$$\text{Labour constraint} \quad L_b + L_g + L_S = 1$$

$$\text{Carbon constraint} \quad \mathcal{E}(T) < \bar{\mathcal{E}}$$

3. RESULTS

I. CHARACTERIZATION OF THE INSTRUMENT

1. Sufficiency-enabling infrastructure is welfare improving if goods are gross complements ($\sigma < 1$)
2. Infrastructure creates a sufficiency channel of structural change

$$\frac{\dot{\omega}_b}{\omega_b} = \underbrace{(1 - \sigma)\omega_g(\varepsilon_b - \varepsilon_g)\frac{\dot{\psi}}{\psi}}_{\text{Income (Engel)}} + \underbrace{(1 - \sigma)\omega_g(g_g - g_b)}_{\text{Price (Baumol)}}$$

With mitigation further constraining sectoral recomposition in the binding case

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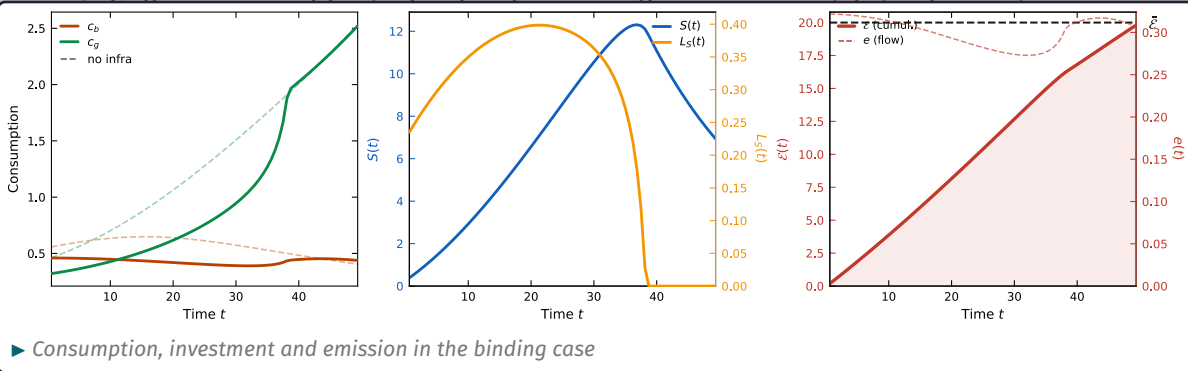
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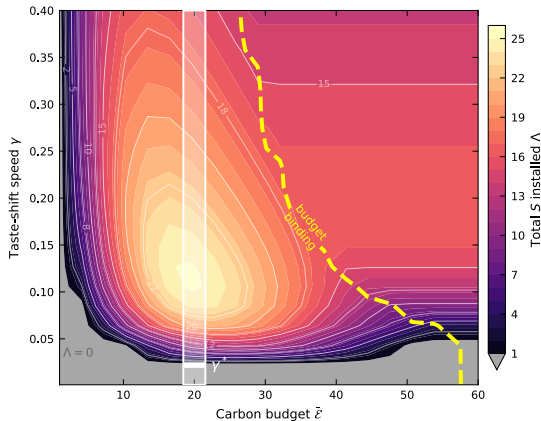
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I. CHARACTERIZATION OF THE INSTRUMENT



II. INVESTMENT DYNAMICS

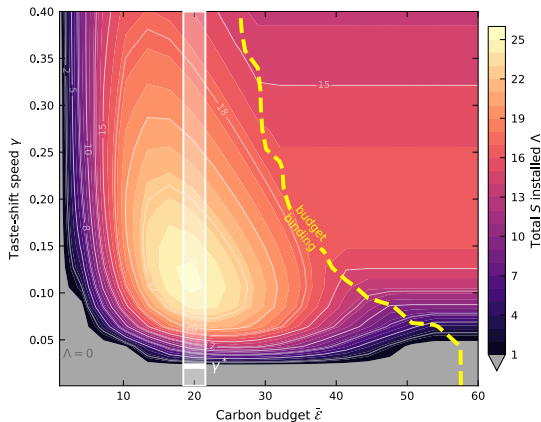
3. Investment starts at $\gamma \geq \gamma^*$
4. Total infrastructure is bell-shaped curved in γ
5. Infrastructure and carbon pricing are complementary instruments
6. Invest first in sectors that are
 - Large and carbon-intensive
 - Easy to taste-shift
 - Cheap and long-lasting in infrastructure
7. Mind the macroeconomic rebound effect



► Total infrastructure as a function of $\bar{E}$ and γ

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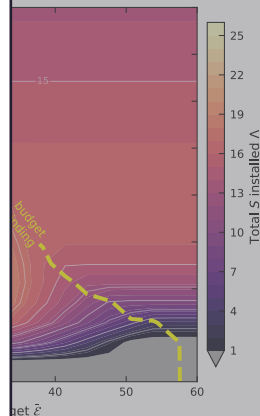
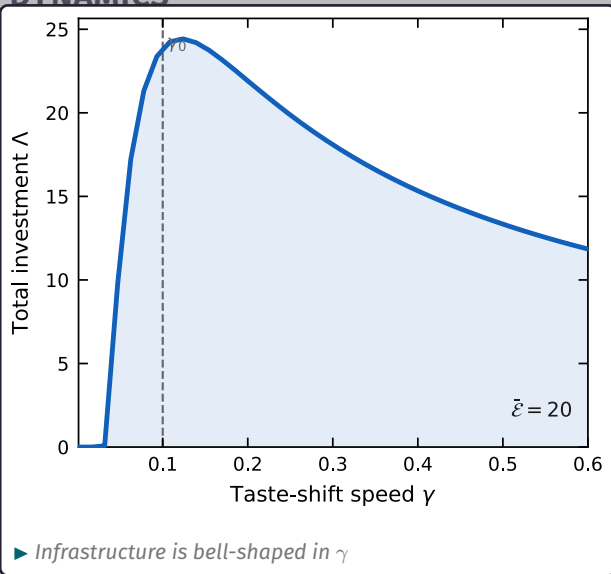
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► Total infrastructure as a function of $\bar{E}$ and γ

II. INVESTMENT DYNAMICS

3. Investment starts at zero
4. Total infrastructure investment is bell-shaped in γ
5. Infrastructure and consumption are complementary
6. Invest first in small γ
 - Large and costly
 - Easy to taste
 - Cheap and low quality infrastructure
7. Mind the macro effect



function of $\bar{E}$ and γ



II. INVESTMENT DYNAMICS

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 6. Invest first in sectors that are
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- Limiting possibilities of carbon pricing raises infrastructure installed
 - Infrastructure allows reaching budget at a strictly lower peak tax



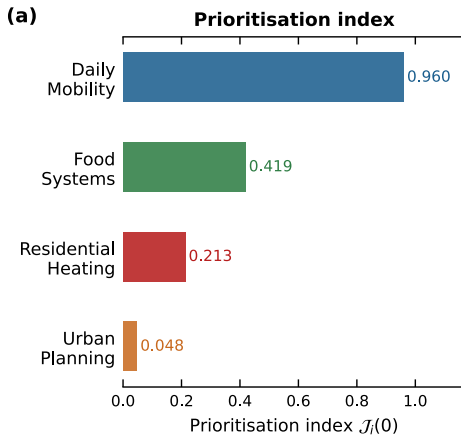
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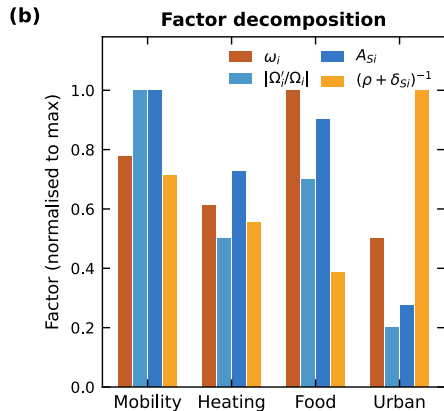
$$\mathcal{J}_i = \frac{\omega_i |\Omega'_i| / \Omega_i A_{Si}}{\rho + \delta_{Si}}$$

Sector prioritisation index

II. INVESTMENT DYNAMICS

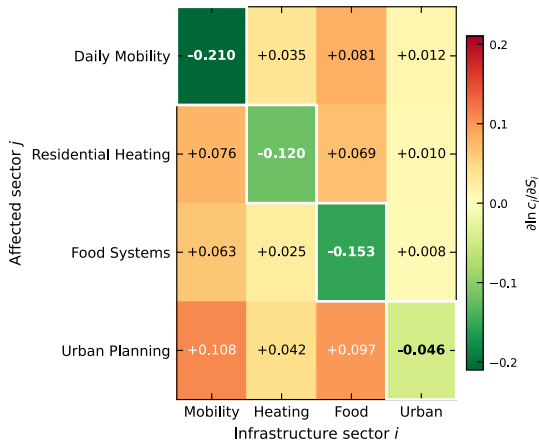


► *Prioritisation index for four fictitious sectors*



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► Consumption reallocation following investment dS_j



4. CONCLUSION

CONCLUSION

This paper

Multi-sector
growth model

1

Infrastructure shapes
preferences endogenously

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carbon budget

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Sufficiency not as an exogenous shock to demand but as the outcome of provisioning systems that policy can deliberately reshape.

Limits

- No distributional analysis
- Limits in empirical validation
S and sector identification
- Interaction with efficiency
- Carbon budget says nothing about feasibility



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Nogent-sur-Marne, June 09. 2026

